

Change Detection in Angular

What it is, how it works, why
OnPush matters

Swipe to master CD like a pro >

What is Change Detection?

Angular uses change detection to update the UI whenever your component's data changes. It checks if component inputs, variables, or bindings have changed – and re-renders the view.

Default Strategy

🔧 Default CD strategy = Checks entire component tree on every event:

- ✓ User input
- ✓ HTTP response
- ✓ Timer / setTimeout
- ✓ Anything in the zone.js lifecycle

➡ Result: Safe but not performant for large apps.

ChangeDetectionStrategy.OnPush



With OnPush, Angular only checks components when:



✓ An `@Input()` reference changes



✓ Observable emits



✓ Manual `markForCheck()` is called

Why it matters:

Massive performance boost in complex UIs

Common Interview Qs

- ? What triggers CD in Angular?
- ? When would you use OnPush?
- ? How do Set or Array.push() affect OnPush detection?
- ? Can CD be triggered without user action?



Be ready to explain with examples!

Real-World Example

```
@Component({  
  selector: 'my-list',  
  changeDetection:  
    ChangeDetectionStrategy.OnPush  
})  
export class MyListComponent {  
  @Input() items: string[] = [];  
  
  addItem() {  
    //  Won't trigger CD – mutation  
    this.items.push('new');  
  
    //  Will trigger CD – new  
    // reference  
    this.items = [...this.items,  
      'new'];  
  }  
}
```